Radiologic evaluation of postoperative gastropericardial fistula.

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Laparoscopic Nissen fundoplication is the current standard surgical option for complicated GERD and symptomatic hiatal hernia. Though comparable in safety, short-term efficacy, and patient satisfaction when compared with open operation, laparoscopic Nissen fundoplication has demonstrated shorter hospital stays and recuperative times. Commonly reported complications include gastric or esophageal injury, splenic injury, pneumothorax, bleeding, pneumonia, fever, wound infections, and dysphagia. We present an unusual case of gastropericardial fistula that developed as a late complication of laparoscopic Nissen fundoplication performed 4 years earlier.

Splenic torsion requiring splenectomy six years following laparoscopic Nissen fundoplication.

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Laparoscopic Nissen fundoplication has become a mainstay in the surgical treatment of gastroesophageal reflux disease, as it has proved to be a durable, well-tolerated procedure. Despite the safety and efficacy associated with this procedure, surgeons performing this advanced laparoscopic surgery should be well versed in the potential intraoperative and postoperative complications. A case is presented of a rare complication of splenic torsion following laparoscopic Nissen fundoplication. Diagnostic evaluations and intraoperative findings are discussed. We present an otherwise healthy 41-year-old woman who underwent a laparoscopic Nissen fundoplication 6 years earlier at another medical center and presented with worsening chronic left upper quadrant abdominal pain. She was diagnosed with torsion of the splenic vascular pedicle, resulting in heterogenicity of perfusion with associated hematoma requiring open splenectomy. Surgeons should be aware of splenic torsion as a potential, albeit rare, complication related to laparoscopic Nissen fundoplication.

A Modified Technique to Create a Standardized Floppy Nissen Fundoplication Without a Bougie.

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Nissen fundoplication is frequently applied in the surgical treatment of patients with gastroesophageal reflux disease (GERD). When the gastroesophageal junction remains too large or becomes too narrow, persistent GERD or dysphagia may occur. To assure a correct size of the gastroesophageal junction, the fundoplication can be created over a bougie. However, this increases the risk of esophageal perforation. Therefore, we have modified a previously described technique to create a standardized fundoplication without the use of a bougie. In this article, we describe this technique and demonstrate the initial results. We describe a technique to create a standardized Nissen fundoplication. After suture repair of the hiatal hernia, three marking sutures were placed on the gastric fundus, based on an equilateral triangle. The size of this triangle determines the final diameter of the fundoplication. With these measurements, we assure sufficient patency, minimize rotation, and create a more reproducible fundoplication that may reduce postoperative dysphagia. We have operated 15 patients according to this technique. Mean operative time was 69.5 (SD 8.4) minutes, no complications occurred. There was no early dysphagia and

the mean length of stay was 1.3 days (1-2). Quality of life after 1 year was excellent. This modified method for standardized Nissen fundoplication is safe and might reduce postoperative dysphagia. Quality of life after 1 year is excellent. The effect on postoperative dysphagia and the reproducibility of this technique should be established in a large prospective study.

Tailored antireflux surgery for gastroesophageal reflux disease: effectiveness and risk of postoperative dysphagia.

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The Nissen fundoplication is not the proper antireflux procedure for patients with poor esophageal peristalsis as it does not strengthen impaired esophageal peristalsis. The aim of this study was to investigate if tailoring of antireflux surgery according to esophageal contractility is an effective treatment of gastroesophageal reflux disease (GERD) with a low incidence of postoperative dysphagia. The Toupet fundoplication was laparoscopically performed on 32 patients with poor esophageal peristalsis and the Nissen fundoplication on 17 patients with normal peristalsis. After a median follow-up of 15 months, only 1 of the 49 patients (2.04%) complained of heartburn. Acute esophagitis was found in none of them on endoscopy. Of 40 patients tested postoperatively, 2 (5%) underwent pathologic esophageal pH monitoring. Postoperative dysphagia was found in two patients (4.1%) compared with 25 (51%) preoperatively ($p < 0.05$). There was a significant reduction of dysphagia following the Toupet fundoplication. Both procedures increased the resting pressure of the lower esophageal sphincter (LES) significantly, which was more pronounced following the Nissen fundoplication. Relaxation of the LES was significantly better following the Toupet than after the Nissen fundoplication. There was significant improvement of esophageal peristalsis following the Toupet fundoplication. Tailored antireflux surgery is an effective strategy for treatment of GERD. The incidence of postoperative dysphagia is low owing to improvement of impaired esophageal peristalsis following the Toupet fundoplication. It may be due to the fact that the Toupet fundoplication causes less esophageal outflow resistance than the Nissen fundoplication.

Minimally invasive surgical techniques in reoperative surgery for gastroesophageal reflux disease in infants and children.

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Fundoplication is commonly performed in children suffering from complications of gastroesophageal reflux disease (GERD). Recently laparoscopic fundoplication has become a standard of care for GERD in children. Published reports show that 2.3 to 14 per cent of children require reoperation after failed fundoplication. The purpose of this study is to show the feasibility of minimally invasive surgical (MIS) techniques to treat children after failed fundoplication. A retrospective chart review was performed for all patients who underwent laparoscopic redo fundoplication at Children's Healthcare of Atlanta at Egleston from July 1998 to July 2000. The patients' records were reviewed for age, diagnosis, type and time of initial operation, type and time of redo operation, operative time for redo operation, and complications. Seventeen children (age 3 months to 18 years) had operations for failed fundoplication attempted using MIS techniques. Six of these children were referred after their initial operation performed elsewhere. Nine (53%) were neurologically impaired. Ten (59%) have respiratory complications of GERD. The initial procedures were as follows: One open Nissen fundoplication, two open Thal fundoplications, 13 laparoscopic Nissen fundoplications, and one laparoscopic Toupet fundoplication. The reoperative procedures performed were revision of fundoplication and hiatal hernia repair (13) or hiatal hernia repair only (four). Two patients had concurrent gastric emptying procedures. One procedure was converted to open for technical reasons. One patient developed a pelvic abscess secondary to leakage around the gastrostomy tube. One child had erosion into the esophagus of a Dacron patch that was used to close a large hiatal defect. Thirteen patients began feeding by the first postoperative day. We conclude that MIS techniques can be applied to reoperative surgery for the treatment of GERD with an acceptable complication rate in this difficult group of patients. Reoperative patients appear to have the same benefits from MIS as patients undergoing their initial procedure.

[Endoscopic surgery for benign esophageal diseases].

PMID: 16910531 | Indexed in PubMed

Gastroesophageal reflux disease (GERD) and esophageal achalasia are common benign esophageal diseases. Today minimally invasive surgery is recommended to treat these diseases. Surgical indications for GERD are failure of medical management, medical complications attributable to a large hiatal hernia, 'atypical' symptoms (asthma, hoarseness, cough, chest pain, aspiration), etc. according to the Society of American Gastrointestinal Endoscopic Surgeons (SAGES) guidelines. Laparoscopic Nissen fundoplication has emerged as the most widely accepted procedure for GERD patients with normal esophageal motility. Partial fundoplication (e.g., Toupet fundoplication) is also considered to decrease the possibility of postoperative dysphagia. Although pneumatic dilatation has been the first line treatment for esophageal achalasia, laparoscopic Heller myotomy and partial fundoplication (e.g., Dor fundoplication) to prevent reflux is preferred by most gastroenterologists and surgeons as the primary treatment modality. Laparoscopic surgery for GERD and esophageal achalasia are effective in most patients and safe in all patients. Finally, laparoscopic surgery should be performed only by skilled surgeons.

Partial fundoplications for gastroesophageal reflux disease: indications and current status.

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The Nissen fundoplication, and in particular the laparoscopic Nissen fundoplication, has received widespread acceptance as the most definitive therapy for gastroesophageal reflux disease. There remains, however, certain patients who do better with a less aggressive surgical augmentation of the lower esophageal sphincter. Partial fundoplications originated in the early 1960s as an alternative procedure to the Nissen, which was associated with moderately high rates of postoperative side effects. These "more physiologic" procedures have proved successful in the treatment of reflux disease in patients with poor or no esophageal motility. In particular, the use of partial fundoplications in association with Heller's myotomy for achalasia has been demonstrated to be well tolerated and to reduce the risk of late dysphasia resulting from uncontrolled gastroesophageal reflux (GER). The use of partial fundoplications in GER patients with normal motility, however, has been less successful. High recurrence rates are documented by many centers with the main cause appearing to be related to a less competent neo-lower esophageal sphincter and a higher rate of wrap herniation. This has led to the current practice of a "tailored approach" to reflux disease, in which all patients receive a thorough preoperative physiologic evaluation to determine the best antireflux procedure for the individual. This is generally a Nissen repair for those with normal motility and either an extrashort "floppy" Nissen or a partial wrap for those with impaired peristalsis.

Intrathoracic herniation and perforation 18 years after open Nissen fundoplication.

PMID: 11071409 | Indexed in PubMed

Nissen fundoplication is the most commonly performed surgical procedure in the management of gastroesophageal reflux disease. Esophageal and gastric perforations most commonly occur in the perioperative period and carry significant morbidity. We describe a unique case of intrathoracic gastric wrap perforation and its suspected pathophysiology almost two decades after the original procedure.

What did the laparoscopic Nissen approach of the gastro-oesophageal reflux really change for the patients 8 years later?

PMID: 11396054 | Indexed in PubMed

Nissen fundoplication (NF) is recognized as the surgical treatment of the gastro-oesophageal reflux disease (GERD). NF can be achieved either by open surgery or by laparoscopic approach. From 1987 to 1997, 210 patients were treated for GERD by NF: 61 by open and 149 by laparoscopic approach. All the patients were followed more than 1 year and were scored by clinical assessment (Visick scale adaptation). In case of Visick score > 1, GI-endoscopy, X-ray series or 24-hour pH-study complete the evaluation. The operative time was comparable between both groups. The postoperative recovery was statistically faster in the laparoscopic group ($p = 0.0001$). The mean time of follow-up was 6 years after open NF and 4 years after laparoscopic NF. After open NF or laparoscopic NF, 72% and 67% of the patients are respectively scored Visick 1, 13% and 21%--Visick 2, 6.8% and 6%--Visick 3 and 8.2% and 6%--Visick 4 (NS). Patients with recurrence of GERD were scored Visick 4, so failure of the surgical treatment is observed in 5 patients after open NF and 9 patients after laparoscopic NF. The occurrence of incisional hernia was significantly higher in the open group ($p = 0.0001$). NF remains a safe procedure for surgical treatment of GERD and can be achieved by laparoscopic approach with comparable results to those by open laparotomy. In our experience, the advantages of the laparoscopic approach is a faster postoperative recovery and a lower risk of incisional hernia.

360 degrees laparoscopic fundoplication with tension-free hiatoplasty in the treatment of symptomatic gastroesophageal reflux disease.

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Since laparoscopic Nissen fundoplication was first described by Cuschieri in 1989 and later by Dallemagne in 1991, this procedure has been widely employed for the treatment of symptomatic gastroesophageal reflux disease (GERD) and/or hiatal hernia. However, a relatively high incidence (7-11%) of intrathoracic Nissen valve migration/paraesophageal hernia following laparoscopic fundoplication has recently been reported. Between November 1992 and August 1995, 65 consecutive patients with severe GERD and/or hiatal hernia underwent laparoscopic 360 degrees fundoplication. In nine of these 65 (13.8%) patients, an intrathoracic Nissen valve migration had occurred within 4 months. Six of these patients were symptomatic and were again submitted to the laparoscopic intervention. Videotapes of both the first and second operation were reviewed. In all cases, it was apparent that, at the first operation, closure by stitches of the hiatus was under tension, and at the second operation, the muscle fibers of the right crus were disrupted, probably due to the tension between the suture margins during the inspiratory movements of the diaphragm. These findings prompted us to perform an effective tension-free closure of the hiatus. A polypropylene mesh (3 x 4 cm) was placed on the hiatus behind the esophagus and fixed with eight metallic agraphes (2 + 2 on the superior edge and 2 + 2 on the lateral sides of the right and left cruses). Between August 1995 and February 1998, the technique, complete with 360 degrees fundoplication, was used for 67 patients with GERD. At mean follow-up of 22.5 months (range, 1-30), there was no evidence of postoperative paraesophageal hernia or complications related to the use of the mesh. This tension-free hiatoplasty seems to be an effective solution to prevent postoperative paraesophageal hernia in patients undergoing antireflux laparoscopic surgery. However, longer follow-up is still needed.